Literature Review

<It is expected that each team member will read and summarize **three new pieces of literature with each new submission!** Keep track of the total number below by incrementing the header. You can choose from peer-reviewed, scientific articles (https://scholar.google.com/), patents (https://patents.google.com/), engineering standards (request from Dr. Arena ahead of time if not accessible through VT or ANSI), or FDA guidance documents (https://www.fda.gov/regulatory-information/search-fda-guidance-documents#guidancesearch). **Do NOT use literature that was separately assigned for homework**.

Note: AI should serve as a collaborator, not a replacement, for your own critical thought and skills. The intellectual contribution and design decisions must remain your own. AI tools may lack direct access to Google Scholar/Patents and the most recent literature. Additionally, AI-generated content can be inaccurate, biased, or incomplete. Therefore, it is up to you to identify the key pieces of literature for your project. AI can be used to support the Summary of Key Findings section, but you should verify the accuracy of the content and that nothing from a piece of literature was overlooked. Please cite the AI tool used, the prompt that you provided, and your analysis of, or updates to, the content. The other sections should be completed without the use of AI.>

1.Risk Management Status of Waste Anesthetic Gas Using ECRI Institute Standards

Source Authors, Journal, Date, Affiliation
This article was published in November of 2012 in the Iranian Journal of Public Health by S. Asefzadeh, AR. Raeisi, and A. Mousavi. I found it on Google Scholar.

Summary of Key Findings
This study was created to assess the risk management of anesthetic waste gas in Iranian academic hospitals to reduce the amount of harm done on employees by the waste gas. The Emergency Care Research Institute (ECRI) created a checklist in 2011 meant to act as a risk assessment of waste gas. The list was made up of 36 questions that assesses the hospital based on planning, training, equipment, evaluation, and monitoring practices of the hospital [1]. The results of the study showed risk management in academic hospitals in Iran was low, and their employees were being exposed to gas [1]. The lowest levels of risk assessment were in planning, training, evaluation, and monitoring. It was found that waste gas exhaled from the patient is not well controlled along with equipment like tubing and connectors that contain the gas [1].

Application to Design Project
It is important to understand where the employees have knowledge of how to contain anesthetic gas and how to assess if there is too much waste gas in the room. From the study we learned that while having a scavenging system is not enough to rid waste gas from the open air, there is also a weakness in training personnel to ensure the equipment is properly secure and how to monitor if there are any issues. There should be training modules to ensure the safety of medical personnel and broaden knowledge of waste gas and the procedures to take when it is leaking.

Specific Information for the Design project
- Find the list of 36 questions to assess our ORs
- Ask for make and model of equipment

2.Three-gas Detection System with IR Optical Sensor Based on NDIR Technology

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Source Authors, Journal, Date, Affiliation
This article was published in November 2015 by Elsevier under the Optics and Lasers Engineering section. The authors of this article are Qiulin Tan, Licheng Tang, Mingliang Yang, Chenyang Xue, Wendong Zhang, Jun Liu, and Jijun Xiong. I found it through Google Scholar.

Summary of Key Findings
In this study, they created a three-gas detection system based on non-dispersive infra-red (NDIR) technique [2]. The principle used in Beer-Lamberts Law I=I0*exp(-KCL), where C is the gas concentration, I and I0 are the light intensity, K is the absorption coefficient, and L is the length of the cell. The idea is that the infrared light passes through nonsymmetric diatomic and polyatomic molecule gases, the molecules will absorb the light energy [2]. In this experiment they measured methane, carbon dioxide, and carbon monoxide. The device contained an infrared source and four single channel pyroelectric sensors [2]. The inside of the chamber was coated with a gold film. In the results, they were able to detect methane from 0-50,000ppm, carbon monoxide from 0-44,500ppm, and carbon dioxide from 0-48,000ppm with an accuracy of +-0.05% [2].

Application to Design Project
If we were to use a sensor to detect anesthetic waste gas for our project, we could use an NDIR sensor and find the concentration of it in the open air. This article uses a different type of gas than anesthetics, but if we can find the wavelength that waste gas can be detected on then we can use this equation. In this experiment they compensate for external factors like extra light seeping through, but I believe we would have to pay for the article to find out how they did that but it would be very helpful.

Specific Information for the Design project
- I=I0*exp(-KCL)
- Non-dispersive infrared sensor
- Gold film inside chamber
- Find out compensation for external factors

3. Establishing Environmental Sustainability in a Resource-Limited Surgical Setting: The Design of Kyabirwa Surgical Center

Source Authors, Journal, Date, Affiliation
This article was published in June of 2025 in the Surgical Endoscopy journal. The authors are Grace Travers, Saul Kibirango, Linda Zhang, Anna Kalumuna, Winfred Nannozi, Ronard Tusiime, Joseph Damoi, Angellica Giibwa, Reetwan Bandyopadhyay, George George, and Michael Marin. I found it on Google Scholar.

Summary of Key Findings
Low- and Middle-class countries are more vulnerable to environmental degradation through climate change because they have weaker infrastructure and a lack of resources [3]. This study highlights ways to be environmentally sustainable in rural medical areas, specifically Kyabirwa, Uganda. Kyabirwa Surgical Center was designed to be environmentally sustainable. Some of the ways they did this were using local materials and natural resources like natural sunlight and natural ventilation to reduce energy used and use environmentally conscious anesthetics [3]. They also prioritized being independent from the local power grid through using solar power and rainwater supply. Through these actions, they found

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that Kyabirwa Surgical Center used 0.4% of its power and 25% of its water from the town which saved $6,121 in annual operating costs [3].

Application to Design Project
We are working with a non-profit organization, and while our focus at the moment is to reduce the amount of waste gas in the OR we should also consider how to make that method sustainable. If we can reduce the amount of cost to operate the OR, then they can provide more services to the area they are in. From what we have learned, they operate in a shipping container. If we can "upgrade" that container by introducing natural light and ventilation, then maybe we can reduce the amount of power used.

Specific Information for the Design project
- Natural Sunlight
- Solar Powered
- Natural Ventilation while keeping sterility
- Possibly use rainwater (?)

4. WAG Treatment and CO2 Absorbers: New Technologies for Pollution and Waste Prevention

Source Authors, Journal, Date, Affiliation:
This article was published in 2018 in the (American Society of Anesthesiologists (ASA) Monitor journal. The authors are Brian P. Barrick and Elizabeth A. Snow. I found it through google scholar.

Summary of Key Findings:
Anesthetic waste gas can stay in the atmosphere between 1.1 years to 114 years with sevoflurane being on the lower end and nitrous oxide being on the high end [4]. The three proposed ways of destructing nitrous oxide are oxidation, reduction, and catalytic splitting. Activated charcoal effectively absorbs volatile anesthetics, methanol, and acetone [4]. Charcoal is great for indoor systems without scavenging but does not help with outdoor pollution because it desorbs and releases quickly. Another method is cold trap condensation where it turns the gas into liquid with cooling. A pressure valve reduces energy use and air entertainment.

Application to Design Project:
We plan to make a stand-alone scavenging system and need to find a way to evacuate the waste gas from the open air. These are all methods that are currently in use that we could potentially take from with modifications to our specific gases.

Specific Information for the Design project:
- Cold trap condensation
- Charcoal

5. Comparison of Waste Anesthetic Gases in Operating Rooms With or Without a Scavenging System in a Brazilian University Hospital

Source Authors, Journal, Date, Affiliation:
This article was published May 18, 2017 in the Revista Brasileira de Anesthesiologia journal. The authors are Leandro Gobbo Braz, Jose Reinaldo Cerqueira Braz, Guilherme Aparecido Silva Cavalcante, Katina Meneghetti Souza, Lorena Mendes de Carvalho Lucio, and Mariana Gobbo Braz.

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Summary of Key Findings:
Operating rooms without active scavenging systems have been shown to have adverse health effects, and this study compared the concentrations of inhaled anesthetics between ORs with and without active scavenging systems [5]. Isoflurane and sevoflurane were measured with an infrared analyzer near the respiratory area of the assistant nurse, anesthesiologist, and near the anesthesia station 30 minutes into surgery and 120 minutes into/after surgery. The concentrations in the ORs with no active systems were significantly higher than those with active systems no matter the location of testing [5].

Application to Design Project:
We are planning to make an active standalone scavenging system for the ORs that are currently using passive systems. This article serves as more evidence that an active system is needed and shows data for it. It would be useful to use the sensor they used to measure concentration of gas.

Specific Information for the Design project:

- InfraRan 4-Gas Anesthetic Specific Vapor Analyzer, Wilks Enterprise, USA)
-

Table 1 Residual concentration of isoflurane (Iso) and sevoflurane (Sevo) (mean and standard deviation) in parts per million (ppm) in operating rooms with and without an anesthetic gas scavenging system. Measurements were performed at three different sites at 30 and 120 min from the start of surgery.

Types of operating room	Inhaled anesthetics (ppm)	Measurement sites					
		Nursing assistant		Anesthesiologist		Anesthesia workstation	
		30 min	120 min	30 min	120 min	30 min	120 min
Without scavenging system	Iso	5.6 (1.8)	9.5 (3.4)[b]	6.2 (2.0)	10.3 (3.4)[b]	9.7 (6.5)	13.3 (6.4)[b]
	Sevo	7.5 (4.7)	16.1 (8.9)[b]	8.5 (6.1)	16.7 (8.5)[b]	14.1 (11.0)	20.0 (8.1)[b]
With scavenging system	Iso	1.3 (0.9)[a]	1.1 (0.5)[a]	1.5 (1.1)[a]	1.2 (0.6)[a]	2.4 (1.0)[a]	3.6 (3.3)[a]
	Sevo	2.3 (1.1)[a]	2.9 (1.1)[a,c]	2.6 (1.3)[a]	3.3 (1.9)[a,c]	3.9 (2.9)[a]	4.8 (2.9)[a]

[a] $p < 0.001$ compared with operating rooms without scavenging systems at all measurement sites at 30 min or 120 min.
[b] $p < 0.01$ compared with 30 min in the operating rooms without scavenging system, at all sites.
[c] $p = 0.02$ compared with isoflurane at 120 min in the operating rooms with scavenging system.

6. Waste Gas Scavenging System

Source Authors, Journal, Date, Affiliation:
This article was published January 29, 2023 through Stat Pearls publishing and is affiliated with San Antonio Military Medical Center. The authors are Nick Lahvic and Mark Liu.

Summary of Key Findings:
A waste gas scavenging system is made up of the relief valve, the conducting tube, and the gas receiver/disposal [6]. Active systems can use fans, vacuums, or a venturi design. Passive systems depend on the gas diffusing independently. The acceptable concentration of volatile anesthetics not in the breathing circuit, based on data from 1977, is 2ppm for halogenated agents and 35 ppm for nitrous oxide [6]. Sevoflurane, desflurane, and isoflurane are large greenhouse gases that should not be released to the outside environment in its original form.

Application to Design Project:
This article gives us more information on the parts of a scavenging system and the different ways we can create a scavenging system. It also provides us information on standards we should aim for in how much gas is in the open air. The rest of the article goes to speak about medical effects breathing anesthetic waste gas has on the body, and the environmental effects it has globally. This information can help us create factors for our project and tailor our needs criteria.

Specific Information for the Design project:

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- Parts of WAGD: relief valve, conducting tube, gas receiver and disposal.
- Possibly use fan
- Need tubing, look into durable kinds of tubing
- Aim for no more than 2ppm concentration of gas in OR
- DO NOT release waste gas into open environment

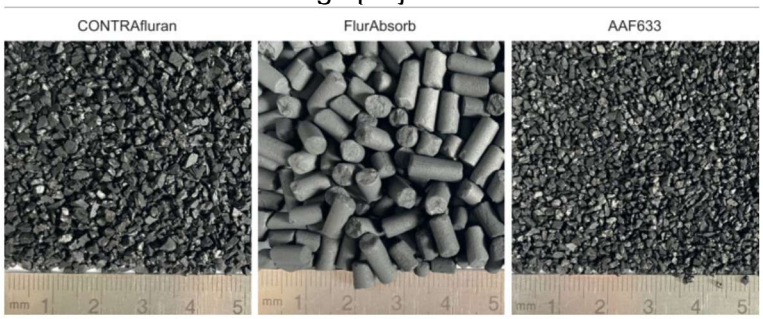
7. Efficiency of passive activated carbon anesthetic gas capturing systems during simulated ventilation

Source Authors, Journal, Date, Affiliation:
This article was published October 4th, 2024 in the British Journal of Anaesthesia. The authors are Lucas M. Muller-Wirtz, Thomas Volk, and Andreas Meiser.

Summary of Key Findings:
This study measured the capturing capabilities of three carbon filter materials for sevoflurane: CONTRAfluran, FlurAbsorb, and AAF633. They found that activated carbon gas filters can sufficiently keep sevoflurane concentration below the occupational exposure limit if the filter capacity and gas flow is maintained well [7]. In this study the critical concentration limit in the air is defined as 10 ppm. The weight of the filter increased linearly with saturation of sevoflurane [7]. The concentration limit was reached at different levels of saturation for each material with the lowest being FlurAbsorb and the highest being AAF633 [7].

Application to Design Project:
We can use this study to help choose what kind of material to use as an adsorber. It seems a material with smaller pellets captures sevoflurane better because it has smaller pores, so the gas doesn't easily pass through it when captured. We can also use their idea that the filter gets heavier as saturation increases to measure if our product is efficient and working.

Specific Information for the Design project:



-
- Weight gain is linear to gas saturation

8. Capture of isoflurane from anaesthetised dogs and cats following methadone, medetomidine and ketamine administration

Source Authors, Journal, Date, Affiliation:
This article was published December 26, 2025 in the Veterinary Record journal. The authors are Florence Hillen, David Yates, and Kate White.

Summary of Key Findings:

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This study evaluated the efficiency of VET-can/VET-dock and SageTech Veterinary in capturing isoflurane. They measured how many animal procedures it took to fill one VET-can. It took 30 animal procedures to fill a VET-can with 81% capture efficiency for isoflurane and 4% capture efficiency for water [8]. Each anesthetic period was about 20 minutes. The VET-can was placed between the scavenging port and active scavenging system and prevented leaks through two self-sealing valves [8]. They did this with corrugated tubing connected to the exhaust port.

Application to Design Project:
The study used something called VET-can to capture isoflurane in animal anesthesia procedures. There is not an image of the VET-can but I assume it is a grain-like substance being used that is inside a can. If we can figure out exactly what is inside the VET-can then we can see if there is a low cost version of it and integrate that substance with our adsorption materials.

Specific Information for the Design project:
- VET-can, VET-dock, SageTech
- Made self-sealing valve to mitigate leakage
- 30 animal procedures, 20min/procedure

9. Gas Detection Handbook

Source Authors, Journal, Date, Affiliation:
This book was published in 2022 under the MSA safety company. There are no listed authors.

Summary of Key Findings:
This guide looks at all components of gas detection systems including different categories of gas monitoring, atmospheres, technologies, gas information tables, classes of gases, and sensor placement. There are 3 steps in correctly placing a sensor. First you must analyze where there could be potential leaks in the system, then create a drawing based on that analysis and include the hazard severity of each site [9]. The next step is to evaluate air flow conditions and determine where gas can accumulate [9].

Application to Design Project:
This guide will be helpful when looking at measuring the gas concentration in the filter and in the room. We have to alert the medical personnel when to change the filter and one was to do that is to measure the gas concentration in the filter environment and around the system.

Specific Information for the Design project:
- Pages 143-147
- Calibrate the sensor to the least sensitive gas if measuring multiple gases

Individual Design Updates

<This section is your electronic design notebook. Include sketches, photos, engineering drawings, equations, calculations, tables, graphs, observations, etc. **Focus on your narration of the design process. For example, your discussion of trainings that you complete and the outputs and/or challenges of that training process (e.g., COMSOL tutorial results, CAD training results, etc.).** Be sure to describe each item in detail with explanatory text so that it is easy to understand when reviewing. It is up to you on how to effectively organize the section headers. Include any notes about

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updates to User Needs (UN), Design Inputs (DI), Design Outputs (DO), Design Verification (VER), and Design Validation (VAL). **There should be at least 3 new pages with each new submission**. In select cases, you may paste and circle **individual items** from your Mural or Miro team boards and describe your contribution in the text below. However, most of the content in this section should be your **individual work**. Make sure that the font size is large enough to be legible, including within figures. However, **a majority of the 3 pages (>=1.5 pages) should be your own descriptive text or narration of the design process. Additionally, it is expected that a majority of the content (>= 1/2) is new or expands beyond what was completed as part of other assignments, such as reading reflections and DHF's. If you include items from a prior individual assignment, please note that in bold/red and expand your discussion.**>

Needs Mapping

I highlighted the problems we discovered/were told by Dr. Read and tried to turn those into design inputs.

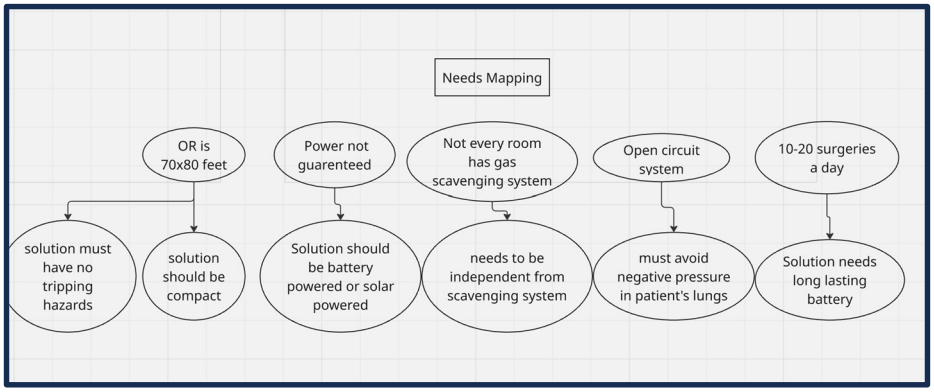


Figure 1. Needs mapping

Solution Mapping

Dr. Read gave us his ideas of a solution, and I wanted to map it out to see if we were missing anything and to see what our possibilities are.

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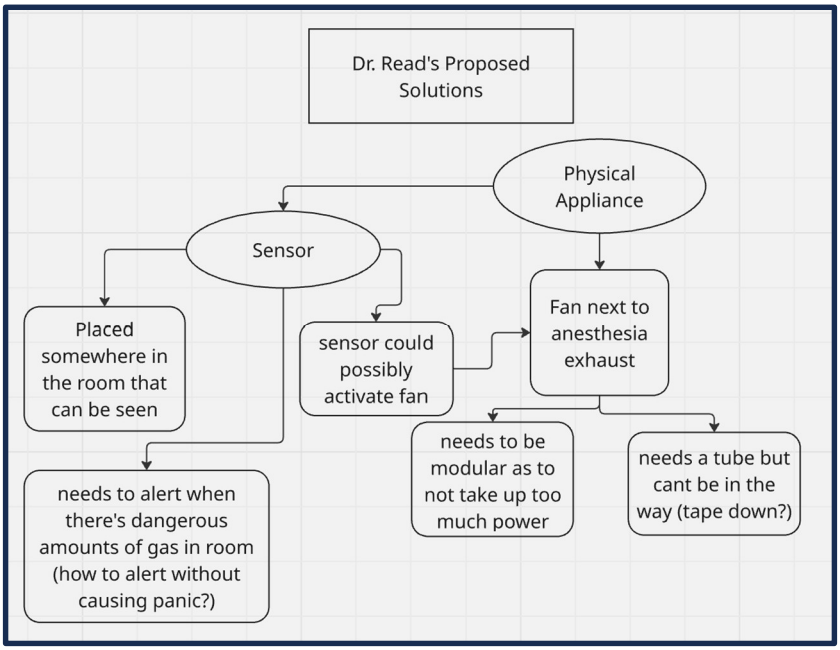


Figure 2. Solution mapping

The original solution we were looking at was a sensor to measure the waste gas and fan it out of the room. After talking with the lead biomedical engineer at Kids OR, Courtney Ludick, we discovered the true need was for an active scavenging system in the operating rooms with only passive systems. She said they have about ten operating rooms with active scavenging systems which are not experiencing drowsiness, but they have 100 operating rooms with passive systems that need to be updated. My though process for developing a solution is to find out first what the current passive system looks like. She said they have a tube running on the floor through the room that outputs into the open air as far from the room as possible. She said the current struggles are that the tubes break easily since they are made of a rubbery material and is being stepped on in the heat of surgery, so it leaks gas. The second thing I wanted to look for was what types of active systems they had and what an active system is made of. Ms. Ludick sent us the manuals for the anesthetic machines used in most of the rooms, so we will have to use those along with the floor plans to help figure out how their active systems are set up. Through the literature review I found that a standard active system is composed of four parts: the relief valve, the conducting tube, and the gas receiver and/or gas disposal. Breaking down each part if the system into components then allows me to look at what exactly we need to research or take into consideration as a user need. Through this though process I have found a few questions I need answered before beginning to form a solution fully: after we trap he gas, do we want to denature it or keep it for disposing without denaturing it? how does the release valve work and what is it made of? Is the conducting tube attached to the patient? What materials can be used as a conducting tube safely?

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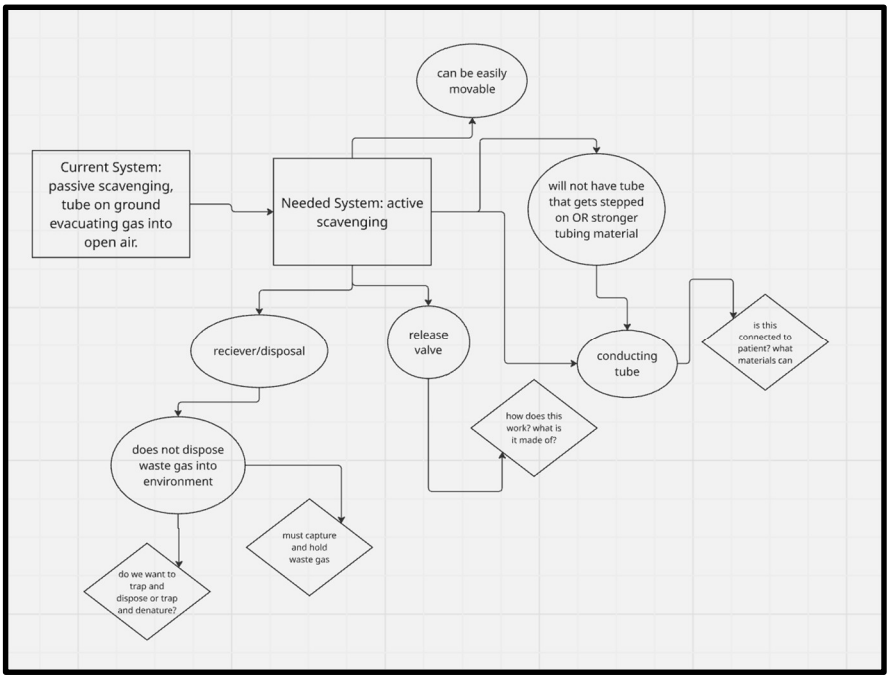


Figure 3. New solution mapping

Sketches of Prototype

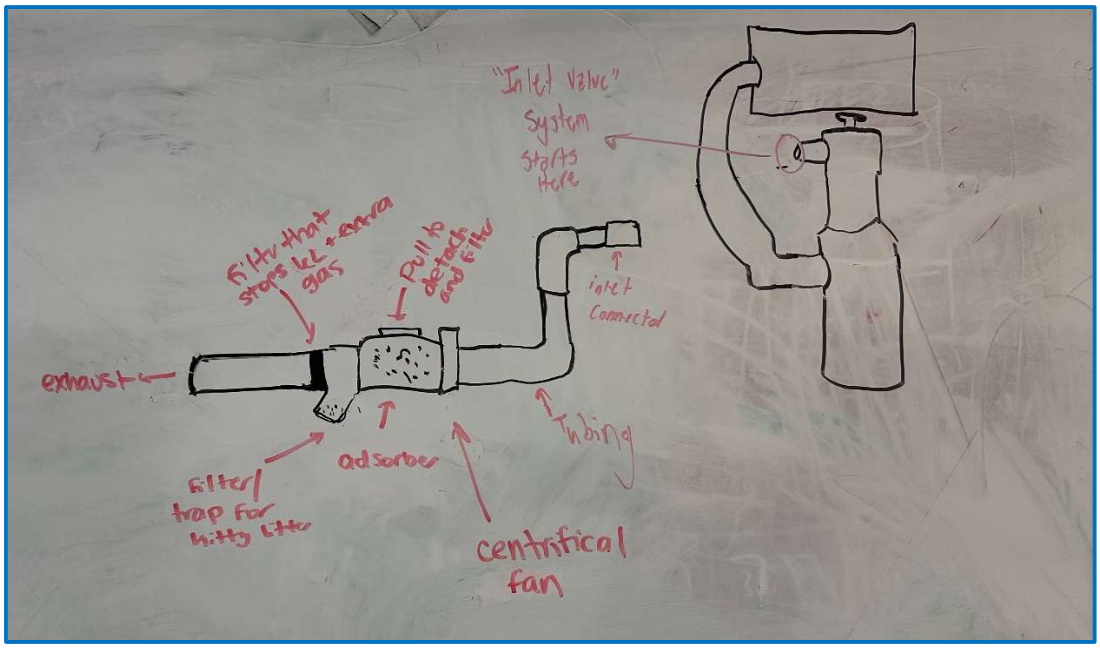
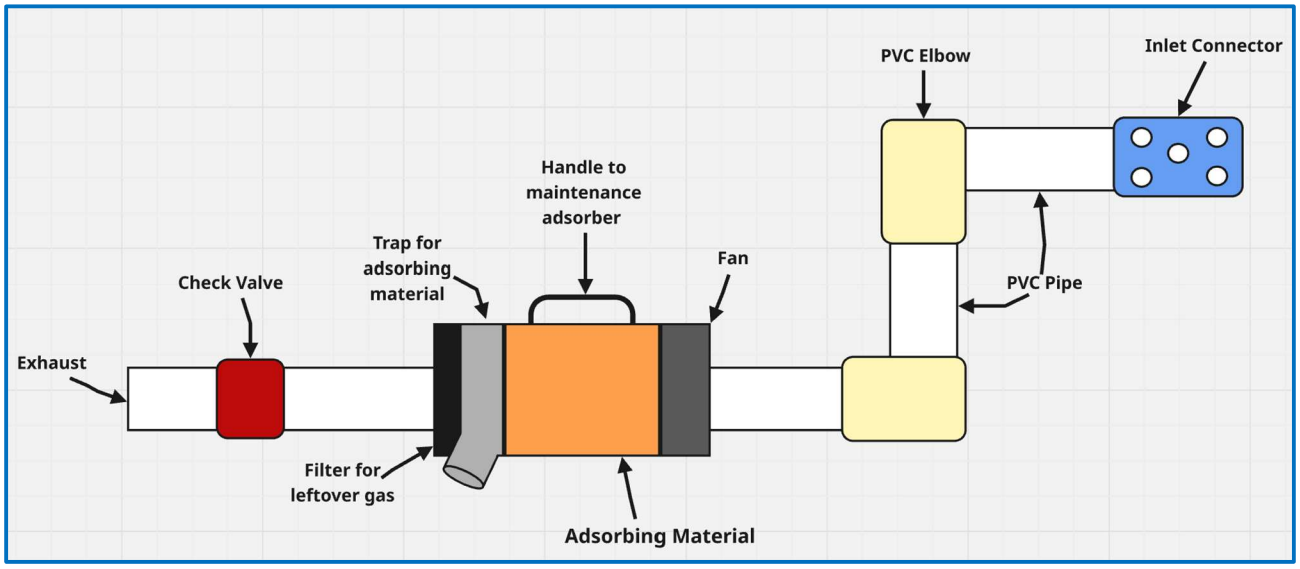


Figure 3. Whiteboard sketch of prototype

Figure 3 is a drawing of our system that we came up with during teamwork time in class. On the far right is a part of the anesthetic machine where we would connect our product to the whole system called the inlet valve. The inlet valve is the end of the built in AGSS transfer system of the anesthetic machine. The first part of our design is the inlet connector. This part will attach to the inlet valve to help

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guide the WAG through the PVC pipes. Because the inlet valve is a couple of inches off the ground, we need to use PVC elbow pipes to connect to the system on the floor. The other components of the system will be too heavy to have a straight horizontal design starting from the inlet valve. The next component is the fan. During teamwork time we discussed using a centrifugal fan to suction the gas through the tube, but after some discussion in our external mentor meeting they said it will be less expensive to use an axial fan. The team still has to discuss which one we want to use. The adsorbing material goes after the fan because the fan is supposed to push the WAG through the adsorbing filter. We are still discussing what material we want to use for the filter because there are so many components to consider. At first, we were looking at kitty litter because of a suggestion Dr. Stone brought up, but our external mentor is not sure it will work because of chemical reactions. This means we have to di further research on a material that will physically adsorb isoflurane and sevoflurane but is also low cost and reusable. Currently I plan to do more research on kitty litter and materials with a sodium bicarbonate base. We have included a handle on the adsorbing system so it can be easier to clean and maintenance. The next part of the system is the trap for the adsorbing material. When we were looking at kitty litter for a solution, we realized that the fan may blow the kitty litter further down the PVC pipe. Now we aren't sure we will be using kitty litter, but any adsorbing material that is in pellet form can be caught in this trap, so we kept it in the design. Next is the filter for excess gas that may not be caught in the adsorber. This part of the design was Sarah's idea; She was concerned that the adsorbing material may not collect all the WAG and that we should implement a safety filter just in case. Initially we were discussing using rubber. This came to be through us searching for a material to use for piping and we thought of rubber. After I did a quick Google search, I realized rubber was a porous material that could potentially adsorb the gases, so we thought maybe it could be used to adsorb leftover gases instead of piping. We brought this idea up to our external mentor, Dr. Read, and he did not like the idea because he doesn't understand how rubber could be used as an adsorbing material. Through his feedback we are still discussing how to use a filter to capture leftover gas. The check valve is a passive system that will be used to ensure there is no backflow in the pipe. We got this idea from Dr. Read and Dr. Andy, and we are still discussing where we should place the check valve. Personally, I think it should be at the end since that is where there is the least amount of flow from the fan so it's more likely to have back flow. The last part of the system is the exhaust. This is where, in theory, clean air will be released to keep a good system flow, and no pressure will build up.



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Figure 4. Electronic sketch of prototype

The difference between the whiteboard sketch and the electronic sketch is labeling and the inlet connector. Our mentors stated that we need to let in air from the OR into the system to prevent negative pressure from our system to the patient's lungs, so I added holes on our inlet connector. In the whiteboard sketch, we have specific materials and components labelled such as a centrifugal fan and kitty litter. In the electronic sketch I made the wording more general since we have not nailed down specific materials yet. Over the next few weeks, we hope to find specific materials based on literature and put them on our ordering list. A way I can make this sketch better is to include specific dimensions of the inlet connector and piping, so I will be working on that over the next few weeks.

Prototype



Figure 5. Design Prototype

Figure 5 is a picture of our prototype that Brooke and Rachel worked on based on my drawing/electronic sketch. The bottom right section is the inlet valve and the section with the blue circles is the inlet connector. The blue circles represent holes in the connector that allow air to flow into the system. Immediately after the inlet connector is the black fan that draws in WAG. The rectangular section with the green foam is the adsorbing material, and the elbow connector next to it is the trap for the adsorbing material. The brown gauze is the excess WAG filter, and the plastic tube covered in gray tape is the check valve. One way we can make this prototype better is to actually drill holes into the outlet, and to separate components with PVC to make the system modular like Dr. Read suggested.

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